GPR learning exercise

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Why I did this

To prepare my application for the EngD on standardizing GPR fieldwork and data management, I wanted to move from a theoretical understanding of GPR to a first practical experience with real data.

What I did

- Processed a real GPR profile (the GSSI example file included in the open-source software GPRPy; a 20.8 m line with a 48 ns time window) in four steps: dewow, background removal, gain, and conversion from time to depth with an assumed wave speed.
- Made a simple simulation of two pipes, 0.6 m and 1.0 m deep, in dry sand and in wet clay.
- Read the file header, first through GPRPy and then directly from the raw file with a short script, and built a metadata form in Excel with 17 fields in five groups: location and time, equipment, ground conditions, processing, and verification.

What I observed

- Processing removed the strong horizontal band near the top (most likely the direct wave between the antennas) and made a sloping layer and deeper reflections much clearer. Processing does not make the radar reach deeper; it makes weak signals from depth visible.
- The processing parameters (the dewow window, the number of traces used for background removal, the gain power and the wave speed) are choices made by the person who processes the data. Changing them changes the image, and a wrong wave speed gives a wrong depth. GPRPy saved all steps and parameters automatically in a history file.
- In my simplified simulation, assumed differences in wave speed and attenuation between dry sand and wet clay made the same pipes appear later and much weaker, and the deeper pipe almost disappeared. The simulation is not a validated physical model, but it illustrates why soil type and moisture need to be recorded.
- Through GPRPy, the header showed only device settings (time window, trace spacing, samples per trace). Reading the raw file directly showed two more fields that GPRPy does not display: the antenna name (“400MHz”) and the date and time the file was created. In the end, 8 of the 17 fields in my form were missing, including all four ground-condition fields, and one was only partly known (the make of the device, but not the model).

What this means for a fieldwork protocol

The metadata in this exercise came from three sources: the device, which wrote its settings into the file; the software, which saved the processing steps; and the operator, whose part – location, surface, soil, moisture and verification – was missing. The exercise also showed that information can be lost between the device and the software even when it is in the file. A protocol should therefore make the operator’s fields a fixed part of every survey, and check that device and processing metadata travel with the data instead of staying behind in the original file. Otherwise, it is not possible to know whether a difference between two surveys comes from the device, the soil or the processing.

What I still need to learn

Converting travel time into depth is simple (depth = speed × time / 2), but the result is only as good as the wave speed. I would like to learn how the speed is best estimated on site, for example from the shape of the hyperbolas or by calibration at a point of known depth, and how its uncertainty could be recorded so that a depth is never reported without it. I also want to gain experience with real GPR equipment, other detection methods, and Dutch standards such as KLIC and IMKL.